Grounding Image Composition with Label-Conditioned Visual Evidence

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Abstract

Image composition labels such as center*,* diagonal*, or* rule of thirds *describe visual structure rather than object categories. KU-PCP provides image-level labels and class-specific geometric elements, but does not show which visible regions people associate with each of its nine classes. We introduce KU-PCP Elements (KU-PCPE), a label-conditioned region-evidence benchmark for photographic composition. Given an image and a label already judged applicable, the task is to segment the visible regions that explain why the label applies. KU-PCPE contains 4,093 single-annotator training masks and a 1,327-task expert test split with three masks per image-label task. The annotations also include exploratory segment-versus-line ratings that describe whether region or semantic-guideline evidence better explains the label. A held-out expert reaches .867 Dice against the other experts, while the strongest deployable model reaches .664, a proposal oracle .672, and the best vision-language model (VLM) proposal selector .395. KU-PCPE exposes the gap between recognizing a composition label and grounding the human-aligned visual evidence that makes it plausible in a particular image.*

1. Introduction

Photographic composition is an inherently visual concept. Labels such as center, diagonal, horizontal, or rule of thirds name recurring spatial organizations, but they do not specify which visible structures make the label applicable in a particular image. Two images may share a label for different reasons, and one image may support several labels through different regions, lines, or partitions. This creates an evidence question that image-level classification cannot answer: given an image and a queried composition label, what visual evidence makes that label plausible?

We introduce KU-PCP Elements (KU-PCPE), a label-conditioned region-evidence grounding benchmark for photographic composition. Each task pairs a photograph with one of its KU-PCP composition labels, and annotators select the visible regions they consider jointly responsible for that composition. A mask may contain one or several disconnected regions and may include objects, object parts, background, or negative space. For labels expressed through a line, axis, or spatial relation, the mask contains the visible elements or partitions that carry the relation. A multi-label photograph therefore yields a distinct grounding task for each of its composition labels.

KU-PCPE builds on KU-PCP but provides a different target. KU-PCP supplies image-level composition labels and class-specific geometric elements such as subject boxes, leading lines, symmetry axes, triangles, and sky regions [10]. KU-PCPE provides one human region representation across all nine classes and three expert interpretations for each test task. A limited comparison with KU-PCP's detected triangles shows shared geometry, while our masks also capture the objects, surfaces, and background that make the composition visible.

KU-PCPE has one primary benchmark: predicting region evidence for an image and an applicable composition label. Class priors, proposal selectors, supervised segmenters, and vision-language model (VLM) selectors are alternative ways to produce the same output, scored against the same expert consensus. Proposal oracles measure how well candidate masks can cover that target. The agreement studies and dataset analyses establish what the target represents and how reliably it can be measured. Modality ratings characterize explanatory form, while classifier heatmaps and deletion experiments connect the masks to image-level recognition. Figure 1 summarizes this organization.

Our contributions are:

- KU-PCPE, a label-conditioned region-evidence dataset with 4,093 training masks and 1,327 three-expert test tasks across nine composition classes;
- a common region target, expert agreement analysis, and studies of label conditioning, annotation workflow, anno-

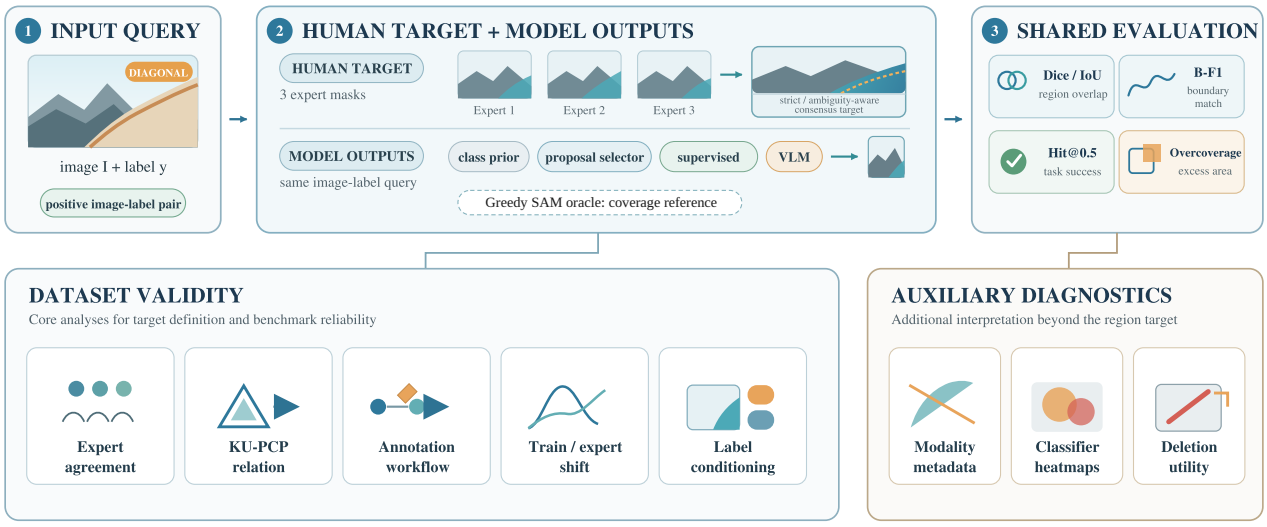


Figure 1. Overview of KU-PCPE. The primary benchmark predicts which visible regions explain an applicable composition label in a given image. Different method families share the target and evaluation. The remaining results either establish the validity of the dataset and target or provide supporting diagnostics of what the masks capture.

tator variation, and the relationship to KU-PCP's geometric elements; and
- a region-evidence benchmark with model-free, proposal-based, supervised, and VLM baselines, complemented by recognition diagnostics.

The dataset, evaluation code, and reconstruction tools are available at adadras.github.io/ku-pcpe/.

2. Related Work

Composition labels and assessment. Most computational work treats photographic composition as a global label, score, or embedding. Broad aesthetics datasets such as AVA and AADB provide aesthetic ratings and attributes, while composition-specific resources such as KU-PCP, CADB, LODB, and PICD evaluate composition classes, quality, retrieval, or learned representations [8, 10, 12, 22, 24, 25]. These resources establish useful composition vocabularies and assessment protocols, but they do not provide a common human region target for a queried label. KU-PCPE adds that target at the image-label level.

Geometric composition elements. Several approaches make composition visible through geometric or pattern-based structure. KU-PCP uses class-specific automatic elements, including main-subject boxes, leading lines, a symmetry axis, triangles, and sky regions [10]. Triangle composition has also been modeled with handcrafted line and geometric descriptors for retrieval and feedback [26], while SAMP-Net learns scores over predefined composition partitions [22]. These representations make selected composition cues explicit. KU-PCPE instead asks people to select visible region support using one representation across all nine labels. We compare the overlapping triangle representation directly in Section 5.3.

Semantic lines and layout primitives. Semantic-line detection makes one important composition primitive explicit by detecting visually meaningful lines that separate semantic regions or convey layout structure [5, 7, 9, 23]. Recent detectors improve line selection by modeling candidate ranking, pairwise relations, or globally harmonious line combinations, and have been used for horizon, vanishing-point, symmetry, and composition-based retrieval tasks. Yet line-based representations remain strongest for straight guidelines such as horizons, diagonals, perspective, and symmetry axes. They are less suitable for curved layouts, repeated patterns, negative space, or region-ordering effects. KU-PCPE therefore treats semantic lines as one possible explanatory modality rather than as complete composition evidence.

Visual and textual explanations. Post-hoc visual explanation methods such as Grad-CAM localize classifier evidence, but they explain a trained model's decision rather than human-marked composition evidence for a queried label [15]. Recent aesthetic and composition work has also moved toward language: neural reviewers generate comments, photo-critique datasets pair images with textual feedback, VILA learns aesthetic representations from user

Class	Train tasks	Expert-test tasks
Horizontal	758	211
Vertical	195	66
Diagonal	471	189
Curved	235	92
Triangle	374	171
Rule of thirds	923	125
Center	750	311
Symmetric	232	99
Pattern	155	63
Total	**4,093**	**1,327**

Table 1. Image-label tasks. The single-annotator train split contains 3,169 images and provides training supervision. The expert-test split contains 1,073 images and three masks per task.

Interface statistic	Train	Expert test
Annotations with records	4,093	3,981
No geometry action	51.8%	62.2%
Any geometry action	48.2%	37.8%
Manual tool use	15.1%	17.5%
Split operation	5.7%	6.1%
Median recorded actions	15	8

Table 2. Annotation workflow across all four annotators. Test percentages and medians pool 3,981 annotations from three experts on 1,327 tasks; training has one annotation per task. Manual tool use includes entering, applying, exiting, or undoing refinement. Per-annotator results are in the supplement.

comments, and MICA jointly predicts composition scores and explanations [6, 16, 17, 21]. These works show that scores and labels are insufficient, but textual critiques can say that an image is balanced, centered, or diagonal without localizing the responsible region, partition, or axis. Since composition is inherently spatial, KU-PCPE complements textual explanations with masks showing which regions support a given label. It also records whether annotators prefer regions or lines as explanations.

3. KU-PCP Elements Dataset

3.1. Evidence Target

KU-PCPE presents each task as a KU-PCP image paired with one of its composition labels. The target contains the visible regions an annotator considers jointly responsible for that composition. Evidence may contain one or several disconnected regions and may include objects, object parts, background, or negative space. For compositions expressed through a line, axis, or spatial relation, the mask contains the visible elements or partitions that carry the relation. Binary masks thus serve all nine classes while allowing experts to choose different regions.

The labels are horizontal, vertical, diagonal, curved, triangle, rule of thirds, center, symmetric, and pattern. Multi-label images contribute one task per KU-PCP label, so each label is grounded in its own visible evidence. The supplementary material provides the full class guide and both representative and ambiguous examples for every class.

Class frequencies vary substantially in both splits (Table 1). The training split provides development supervision. Human agreement and benchmark evaluation are defined on the three-expert test split.

3.2. Annotation Protocol

Annotators first saw an image and its queried composition label. Precomputed candidate regions were revealed by clicking in the image (Figure 2a). They could select several regions, add or split regions, remove selections, and manually refine the final mask. The candidates were generated with SAM2 Hiera-L [13]. The candidates reduced the effort of drawing boundaries, while the annotator decided which visible regions supported the label and confirmed the result. Annotators worked independently and did not see one another's masks. They received written instructions and a guide defining all nine classes, with three visual cues per class (see the supplement).

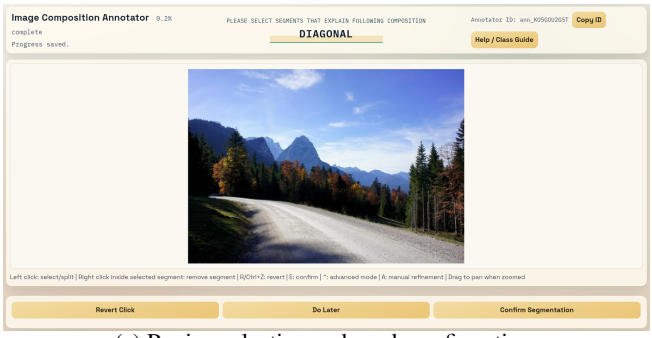
The three test annotators all had formal education in arts or design and more than five years of professional experience in art, design, photography, or professional visual practice. A separate annotator produced the training split with the same task and interface. Records from all four annotators show that most annotations had no geometry action logged (Table 2). Section 5.4 combines the three experts' retained records at the task level.

The interface also displayed automatically detected semantic lines. After confirming a mask, annotators could rate whether the region mask or the line overlay better explained the queried label (Figure 2b). The overlay served as an annotation aid, while the confirmed region mask defined the binary target.

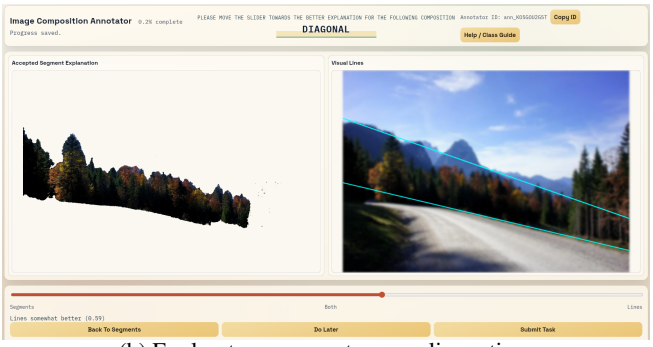
3.3. Expert Agreement

We have three masks for every test image and applicable label. Their pixelwise majority defines the *strict* target. The *ambiguity-aware* target complements masks whose opposite side agrees better with the first expert before taking the majority. This captures partition ambiguities such as selecting sky or ground around the same horizon. Both targets are included in the annotation package and reported.

Mean raw pairwise Dice is .7428. A held-out expert reaches Dice .8666, intersection-over-union (IoU) .8054, boundary F1 .7541, and Hit@0.5 .9312 against the aligned consensus of the other two experts. Table 3 shows strongest agreement for rule of thirds, center, and triangle. Horizontal, diagonal, curved, and pattern admit more complementary or alternative readings, as illustrated in Figure 3.



(a) Region selection and mask confirmation.



(b) Exploratory segment-versus-line rating.

Figure 2. KU-PCPE annotation interface. Annotators select or refine visible region evidence for the queried composition and then compare the confirmed mask with an automatic semantic-line overlay.

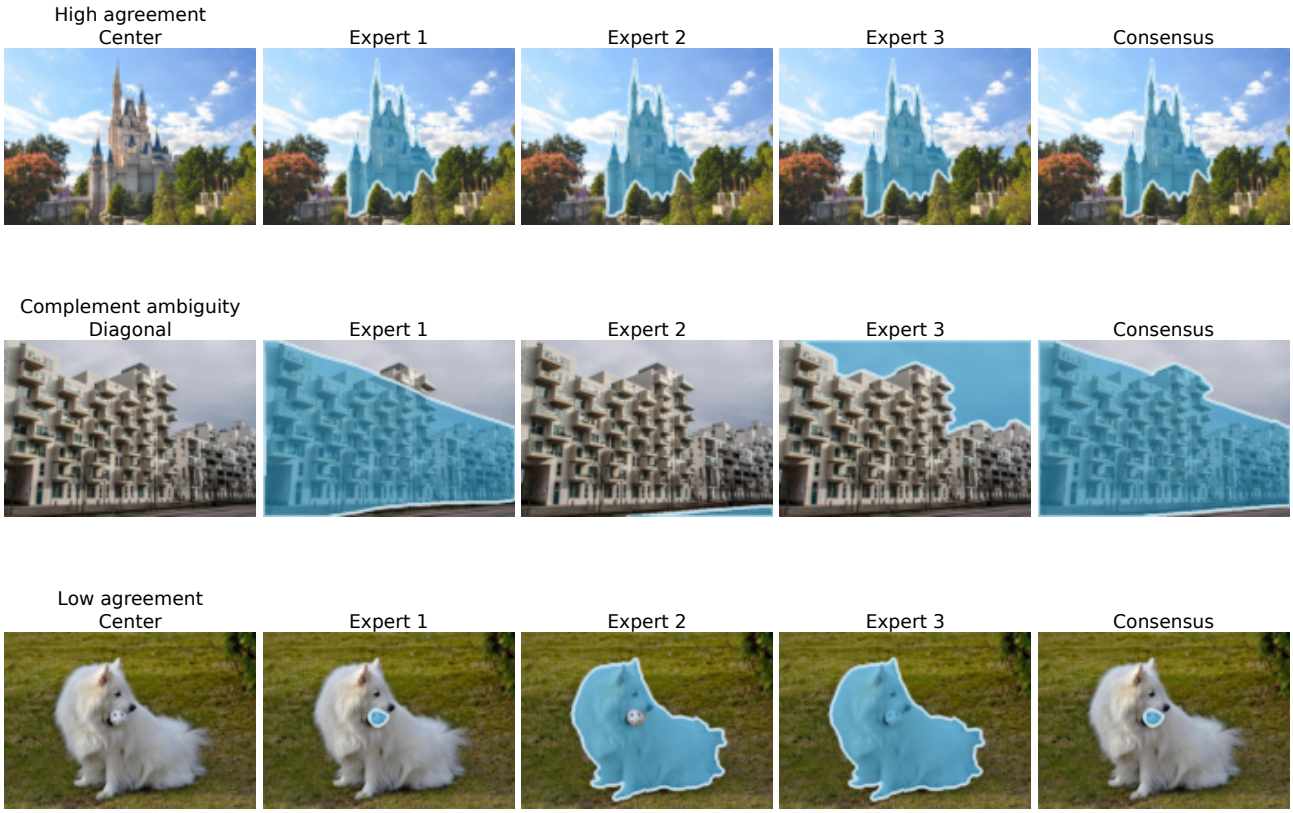


Figure 3. Expert agreement examples. Rows show high agreement, complementary region ambiguity, and low agreement. Columns show the image, three expert masks, and the aligned majority. Disagreement often reflects alternative plausible supports rather than small boundary differences.

3.4. Exploratory Evidence Modality

Experts also rated whether the confirmed region mask or an automatic semantic-line overlay better explained the queried label, using a score $r \in [0, 1]$ from segment to line preference. After binning ratings as segment, mixed, or line, pairwise agreement is .54 and Fleiss $\kappa = .26$; continuous ratings have mean pairwise $|\Delta r| = .22$. These exploratory ratings describe whether experts prefer regions or lines as explanations. The three preferences appear in Figure 4. Horizontal is mainly mixed, diagonal and triangle have the largest line-preferred shares, and most other classes are segment-dominated (Figure 5).

Class	Tasks	Pair D	Maj. D	κ	Flip
Horizontal	211	.6371	.9539	.8281	.5118
Vertical	66	.6557	.8362	.5782	.1515
Diagonal	189	.5973	.8805	.5433	.4656
Curved	92	.6105	.8556	.5240	.3696
Triangle	171	.8332	.9351	.7796	.1170
Rule of thirds	125	.9538	.9782	.9426	.0080
Center	311	.8427	.9235	.8014	.0354
Symmetric	99	.8239	.9165	.7152	.0505
Pattern	63	.5324	.8092	.2510	.5397

Table 3. Expert agreement by class. Pair D is raw pairwise Dice; Maj. D is Dice to the jointly aligned majority, with expert 1 fixed and the remaining orientations maximizing mean pairwise Dice. Flip is the fraction of tasks for which at least one complement alignment is used. Full metrics and confidence intervals are in the supplement.

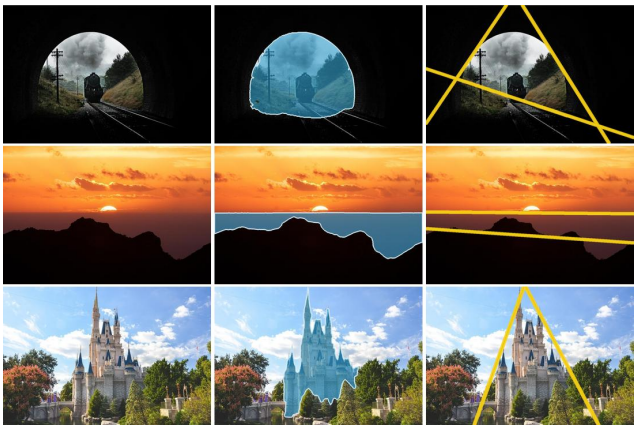


Figure 4. Exploratory modality examples. Top: segment-preferred center ($r = .22$); middle: mixed horizontal ($r = .50$); bottom: line-preferred triangle ($r = .81$). Each row shows the image, human region evidence, and automatic semantic-line overlay.

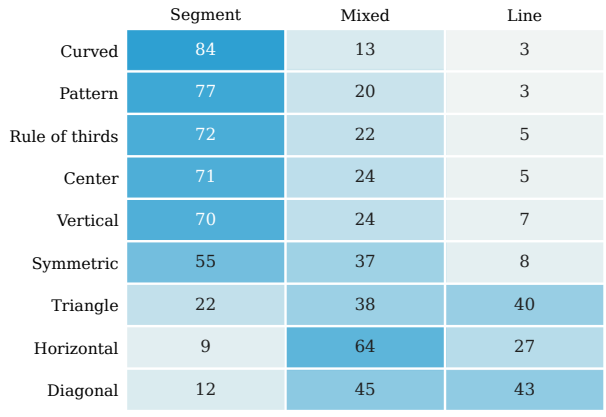


Figure 5. Evidence modality by composition class. Cells give the percentage of expert-test ratings that prefer segments, mixed evidence, or lines. Full values are in the supplement.

4. Benchmark Tasks

Evidence segmentation is the primary benchmark task. The prediction protocols described below differ in how they produce the same binary target; proposal oracles measure candidate coverage, while modality metadata remains exploratory.

Evidence segmentation. Given an image I and a composition label y assigned to that image in KU-PCP, a method predicts a binary evidence mask M. Unless stated otherwise, evaluation uses the ambiguity-aware expert consensus on the test split.

Proposal coverage and selection. SAM2.1 Hiera-L produces a candidate set for each image. Greedy Oracle@k measures how much of the target can be covered when the reference mask is available. Random, area-based, and class-template selectors choose without access to the reference and are deployable controls.

Supervised segmentation. Image-only and label-conditioned U-Net [14] variants and a label-conditioned SegFormer-B0 [19] learn from masks provided by one training annotator; label-conditioned models also receive the queried composition label.

VLM proposal selection. For each task, a VLM sees the queried label and an overlay with up to 12 numbered proposals, then returns up to three IDs. Their union is evaluated as a binary mask. This yields a language-conditioned proposal selector evaluated against the expert region target.

Exploratory modality metadata. Segment-versus-line prediction uses the expert ratings as an auxiliary diagnostic. Full VLM modality results are reported in the supplement.

5. Experiments

We organize the results in three stages. First, we examine the dataset and target through agreement, evaluation sensitivity, the relation to KU-PCP, annotation workflow, annotator variation, and label conditioning. Next, all predictive methods are compared on the same region-evidence task and metrics; proposal oracles report candidate coverage. Finally, classifier explanations and deletion connect the annotations to image-level recognition as auxiliary analyses.

5.1. Benchmark Protocol

The proposal experiments use SAM2.1 Hiera-L. Supervised baselines comprise an image-only U-Net, a label-conditioned U-Net, and a label-conditioned SegFormer-B0. Model-free baselines include random matched-area masks

Method	Strict Dice	Ambig. Dice
Train class template	.451	.462
SAM2.1 Oracle@3	.679	.672
SegFormer-B0	.662	.664
Qwen3-VL-32B	.399	.395

Table 4. Sensitivity to target construction. The benchmark pattern is stable under strict and ambiguity-aware evaluation.

and class templates. We also evaluate a greedy proposal oracle and a deployable template selector. The supplement specifies the training, proposal, prompting, and thresholding settings. The recognition diagnostics use a ResNet-50 composition classifier, Grad-CAM explanations, and DINO ViT patch-similarity maps [2, 4, 15]. The VLM suite covers Qwen3-VL, InternVL3.5, MiniCPM-V 4.5, Molmo2, and LLaVA-OneVision [1, 3, 11, 18, 20]; exact checkpoints are listed in the supplement.

Metrics. Dice and IoU measure region overlap. Boundary F1 (B-F1) uses a two-pixel tolerance after resizing the longer image side to at most 256 pixels. Overcoverage is the fraction of a predicted mask that lies outside the target; lower is better. Hit@0.5 is the fraction of tasks with Dice at least .5. Main results use the ambiguity-aware target. Strict results and 95% bootstrap confidence intervals over image-label tasks are reported in the supplement.

VLM responses. The prompt shows the queried label and up to 12 numbered proposals and asks for a JSON list of at most three IDs. We keep the first three distinct valid IDs; duplicates and unavailable IDs are ignored, and a response with no valid ID produces an empty mask. Decoding is deterministic. Exact prompts, parsing rules, and invalid-response rates are provided in the supplement.

5.2. Evaluation Sensitivity

Sensitivity analysis uses both strict and ambiguity-aware targets. The comparison in Table 4 shows that the central ordering is unchanged. Oracle@3 is selected separately for each target; all other predictions are fixed. Full metrics and confidence intervals are reported in the supplement.

5.3. Relation to Original KU-PCP

KU-PCP's local elements span boxes, lines, axes, triangles, and sky regions. The 171 triangle tasks with matched KU-PCP detector output provide a direct comparison between a geometric element and the KU-PCPE region target. We compare both the detected sides and the image region they enclose with the KU-PCPE targets.

The partial overlap in Table 5 confirms that the geometric triangle and the human region target describe related structure. The human region target additionally captures the objects, surfaces, and background regions through which the triangle is perceived. Details and examples are in the supplement.

Triangle comparison	Strict	Ambig.
Sides inside target	.374	.398
Sides inside target, 5-px tolerance	.465	.489
Enclosed region Dice	.572	.585
Enclosed region IoU	.451	.462

Table 5. Comparison with KU-PCP triangle detector output on 171 matched tasks. The first two rows give the fraction of detected side pixels supported by the expert target; the last two compare the enclosed region.

Annotation	n	Pair D	O@3	T@3	SegF.	Qwen
No edit logged	615	.858	.785	.481	.766	.472
Edit logged	712	.643	.574	.358	.575	.328

Table 6. Agreement and Dice by workflow across all three experts. A task has an edit logged if any expert recorded a geometry action; each task contributes once. Pair D is raw pairwise expert Dice; O@3 and T@3 are Oracle@3 and Template@3. SegF. is SegFormer-B0; Qwen is Qwen3-VL-32B. Predictions and target-aware oracle masks are fixed before grouping.

5.4. Results by Annotation Workflow

We combine the three experts' retained interaction records. A test task is in the no-edit group only when none of its experts recorded a geometry action; it is in the edit group when at least one expert did. Agreement uses all three final masks, and model scores use the ambiguity-aware target. Each of the 1,327 image-label tasks contributes once.

Table 6 shows that tasks with an edit logged have lower expert agreement and lower Dice for all four methods. Agreement, Oracle@3, and SegFormer are lower in every class; Template@3 and Qwen are lower in eight of nine. The exception has only one no-edit symmetric task. Full metrics, confidence intervals, and results by class and individual expert are in the supplement.

5.5. Variation Across Annotations

The training masks and test masks were produced by different people, and the test consensus combines three interpretations. The supplement compares marginal mask statistics on different image splits. These differences combine image content, annotator choices, and consensus construction; they do not isolate an annotator effect.

The training masks are close in area to two experts and smaller than the third, while containing more connected components. Rescoring representative methods against each expert and both consensus targets preserves their central ordering: the greedy oracle measures coverage using the target, while SegFormer-B0 is the strongest deployable method. Using all proposals, the template exceeds the best

Method	Dice↑	IoU↑	B-F1↑	Over.↓	Hit↑
Human reference	.867	.805	.754	.113	.931
Random matched area	.238	.139	.037	.717	.007
Train class template	.462	.325	.169	.481	.494
SAM2.1 Oracle@3	.672	.605	.632	.067	.661
SAM2.1 Template@3	.415	.334	**.447**	.510	.399
U-Net, no label	.496	.379	.307	.458	.534
U-Net + label	.597	.479	.390	.390	.661
SegFormer-B0	**.664**	**.576**	.437	**.300**	**.734**
Qwen3-VL-32B	.395	.324	.440	.523	.376

Table 7. Evidence grounding on the ambiguity-aware target. Human: held-out expert versus aligned peer consensus. Proposal rows use the full bank; Qwen uses 12. Bold marks the best deployable result per column.

VLM, but their Dice scores are similar using the same 12 proposals (Table 8). Training provides supervision; the three-expert test split defines evaluation.

5.6. Label Conditioning

On 244 multi-label images, 264 same-image, different-label task pairs have mean Dice .3655 and IoU .3247. Expert masks therefore depend on the queried label rather than reducing to one generic foreground mask. Pair-level and class-template controls are reported in the supplement.

5.7. Evidence Grounding Baselines

Table 7 and Fig. 6 show the gap between coverage and selection: Oracle@3 reaches .672 Dice, while Template@3 reaches .415. Label conditioning improves U-Net Dice from .496 to .597. SegFormer-B0 nearly matches oracle overlap but remains well below human reliability. On the strict target, Oracle@3, SegFormer-B0, and Qwen3-VL-32B reach .679, .662, and .399 Dice, preserving the same conclusion.

Modality diagnostic. The exploratory segment-versus-line ratings have modest agreement and are strongly class-structured. A class prior estimated from the training ratings reaches .7204 accuracy, while the strongest VLM reaches .4348. This diagnostic characterizes the auxiliary metadata; full VLM results are in the supplement.

5.8. VLM Proposal Selection

For each task, selected proposal IDs are joined and evaluated as a mask. The seven fixed VLMs range from .2910 to .3946 Dice. Table 8 compares the unchanged oracles and template computed from training masks using only the proposals originally shown to each VLM. Of 1,327 tasks, 67 have fewer than 12 proposals. Qwen3-VL-32B and Template@3 now reach .3946 and .4016; the paired Dice difference is $-.0070$ (95% task-bootstrap CI $[-.0216, .0075]$). Thus the full-bank gap partly reflects availability, while both selectors remain well below the exact up-to-three Dice ceiling of .6448. The full-bank greedy reference reaches .6718 versus .6400 on 12; greedy selection is not a guaranteed ceiling. Strict results, all seven VLMs, per-class scores, and uncertainty are in the supplement.

Same 12 candidates	Dice	IoU	B-F1
Template@3	.402	.320	.441
Qwen3-VL-32B	.395	.324	.440
Greedy Oracle@3	.640	.574	.596
Exact, exactly 3	.640	.575	.596
Exact, up to 3	.645	.582	.610

Table 8. Matched-budget selection on the original VLM-visible candidates. All rows use the same ordered bank, with fewer candidates when unavailable. Exact oracles optimize Dice using the target; they are not predictors.

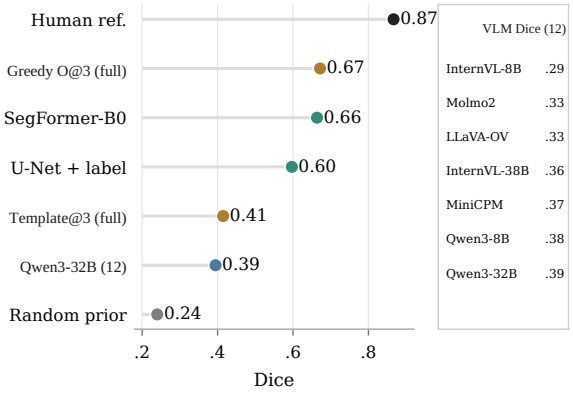


Figure 6. Evidence-grounding performance. Representative Dice scores show the gaps among priors, VLM and template selection, supervised segmentation, proposal coverage, and human agreement. Full-bank proposal methods and 12-candidate VLMs are identified; matched controls are in Table 8. The inset compares seven VLMs using abbreviated names; full names and metrics are in the supplement.

Supplementary examples show supervised successes, errors in selecting proposals, and targets that proposals fail to cover.

5.9. Auxiliary Recognition Analyses

Post-hoc classifier explanations. This auxiliary diagnostic asks whether successful image-level recognition localizes human evidence. Its separately trained classifier is distinct from the supervised region predictors. We train a ResNet-50 on the KU-PCP training labels and evaluate Grad-CAM for the queried class. On KU-PCP test labels, the classifier reaches .9012 mean average precision (mAP), .8215 macro-F1, and .9541 label accuracy at a .5 threshold. We also evaluate DINO ViT patch-similarity maps. The supplement reports area-matched thresholds and the strongest result from a fixed area sweep for both heatmap families.

Mask	Blur Δ	Fill Δ	Blur flip	Fill flip	Dice
Human	1.02	2.03	.10	.21	1.00
Random area	.16	.38	.02	.03	.28
Full image	2.31	4.08	.26	.76	.41
U-Net + label	1.16	2.80	.12	.28	.60
SAM O@3	.55	1.31	.06	.14	.67
SAM T@3	.71	1.75	.07	.19	.41
Qwen3-32B	.62	1.58	.06	.17	.39

Table 9. Counterfactual deletion utility. Human masks have larger effects than random masks of equal area. Full-image and over-complete masks can cause larger drops, so deletion complements rather than replaces human alignment. O@3 denotes Oracle@3 and T@3 denotes Template@3.

Classifier heatmaps, especially Grad-CAM, can overlap expert masks. Low boundary agreement distinguishes recognizing labels from localizing the evidence people select.

Counterfactual deletion. Counterfactual deletion tests whether a mask identifies regions used by the composition classifier. We either blur the masked region or replace it with the image mean, then measure the target-class logit drop and the fraction of predictions that fall below threshold. These perturbations are applied to the classifier described above.

In Table 9, human masks produce much larger drops than random masks of the same area, supporting their functional relevance. Yet full-image and U-Net masks cause larger drops despite weaker human alignment, and Template@3 exceeds the higher-Dice Oracle@3. Regions affecting the classifier and those selected by people are therefore related but not interchangeable.

6. Discussion

KU-PCPE turns a composition category into an image-specific grounding task. Low overlap between different labels on the same image supports a target that depends on the queried composition. The triangle comparison relates geometric structure to the objects, surfaces, and negative space in human region masks.

The results separate proposal coverage, semantic selection, and learning. Oracle@3 reaches .672 Dice, showing that candidate regions often cover the target, while the full-bank template selector and strongest VLM reach .415 and .395. On the same 12 candidates, template and VLM Dice are similar (.402 and .395), below the exact up-to-three ceiling (.645). Supplying the queried label improves U-Net from .496 to .597, and SegFormer reaches .664, close to proposal-oracle overlap but below expert reliability at .867. These gaps place label-conditioned selection and precise region assembly at the center of the task. PICD [25] found that its best specialized model distinguished compositions more accurately than the tested VLMs. Here, supervised segmenters also align more closely with human evidence than VLM selectors. The gap extends from composition discrimination to identifying the visual structure that explains a label. Classifier explanations and deletion further show that recognition-sensitive pixels can differ from the evidence people select.

Expert disagreement is structured by class: rule of thirds, center, and triangle show strong agreement, while horizontal, diagonal, curved, and pattern admit more alternative supports. Method ordering remains stable under strict and ambiguity-aware targets. Individual masks and consensus targets, formed with or without complement alignment, support studies of uncertainty and alternative valid evidence.

Scope and limitations. Candidate regions were used during annotation. Tasks with geometry actions recorded by any of the three experts have lower agreement and lower model scores. The workflow comparison describes an association with task difficulty; it does not isolate candidate quality or the causal effect of editing. All experts used candidate assistance, so shared candidate bias remains unresolved; a proposal-free independent audit has not been performed. The training split reflects one annotator's style and serves as learning supervision; final evaluation uses the three-expert test split.

KU-PCPE grounds the positive labels inherited from KU-PCP. Extending the benchmark to negative queries would require a separate target definition: the absence of symmetry or a triangle may arise without a localized counter-evidence region. A complete system could combine label prediction with KU-PCPE grounding for positive decisions. The segment-versus-line ratings are exploratory, and their modest agreement supports class-level analyses while instance-level modality remains uncertain. The benchmark covers KU-PCP's nine labels and photographic images; new domains would require their own definitions and agreement studies.

7. Conclusion

Composition is inherently visual: its meaning in an image depends on the regions and relations through which people perceive it. KU-PCPE makes those regions explicit for training and evaluation across nine classes, retaining shared judgments and alternative interpretations through individual masks and two consensus views. Specialized segmenters agree more closely with people than the tested VLM selectors, but remain below agreement among experts. The dataset therefore supports comparisons between the visual evidence selected by people and models, providing a step toward understanding how machine learning perceives stylistic attributes and how that perception relates to our own.

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Societal Impact

KU-PCPE masks describe photographic structure rather than identity or sensitive attributes. Original participant identifiers and calendar timestamps are excluded. The complete export includes pseudonymous tool counts and relative timelines grouped into five-second bins. Source photographs are not redistributed; users reconstruct the annotations against an independently obtained KU-PCP copy under its original terms. The annotation study involved adult annotators. Permission covers continued processing and pseudonymized release.

Acknowledgments

This work was supported by the Austrian Science Fund (FWF), project DFH 37-N, “Visual Heritage: Visual Analytics and Computer Vision Meet Cultural Heritage”.